Q1 Prime Number

10 marks

A mathematical analysis system needs to verify whether a given number is prime. A prime number is a number greater than 1 that has no divisors other than 1 and itself.

Write a Python program that:

- Accepts an integer input from the user
- Checks whether the number is prime using appropriate control flow constructs
- Displays a suitable message indicating whether the number is prime or not

Ensure proper use of loops and conditional statements.

Your Code

```
n = int(input("Enter a number: "))

if n <= 1:
    print("Not Prime")
else:
    prime = True

    for i in range(2, n):
        if n % i == 0:
            prime = False
            break

    if prime:
        print("Prime")
```

Input

```
13
12
```

Output

```
Enter a number: Prime
```

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PYTHON PROGRAMMING

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Q2 Fibonacci series

10 marks

A numerical computation module is designed to generate a sequence where each number is the sum of the two preceding numbers. This sequence is known as the Fibonacci series.

Write a Python program that:

- Accepts a number n from the user
- Generates and displays the first n terms of the Fibonacci sequence
- Uses looping constructs to compute the sequence

Ensure the program handles cases where n is less than or equal to zero.

Your Code

```
# Python program to generate the first n terms of the Fibonacci sequenc

n = int(input("Enter the number of terms: "))

if n <= 0:
    print("Please enter a number greater than 0.")
elif n == 1:
    print("Fibonacci Sequence:")
    print(0)
else:
    a = 0
    b = 1
    print("Fibonacci Sequence:")
    for i in range(n):
```

Input

```
7
```

Output

```
Enter the number of terms:
Fibonacci Sequence:
0 1 1 2 3 5 8
```

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Write a Python program that:

• Accepts a string input from the user

• Counts the number of vowels, consonants, digits, and special characters present in the string

• Displays the count for each category

Use appropriate string operations and control flow statements.

Your Code

```
# Python program to count vowels, consonants, digits, and special chara

s = input("Enter a string: ")

vowels = 0
consonants = 0
digits = 0
special = 0

for ch in s:
    if ch.isalpha():
        if ch.lower() in "aeiou":
            vowels += 1
        else:
            consonants += 1
    elif ch.isdigit():
        digits += 1
    else:
        special += 1

```

Input

```
Python123@#
```

Output

```
Enter a string: Vowels = 1
Consonants = 5
Digits = 3
Special Characters = 2
```

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Q4 ODD or EVEN

10 marks

A program is designed to check whether a number entered by the user is even or odd. However, the program contains syntax and logical errors.

Analyze the given code and perform the following:

- Identify all errors present in the program
- Rewrite the corrected program
- Briefly justify each correction

num = input("Enter number: ")

if num % 2 == 0

print("Even")

else

print("Odd")

Your Code

```
num = int(input("Enter number: "))

if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Input

```
8
5
```

Output

```
Enter number: Even
```

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Q5 **Print numbers from 1 to 10** 10 marks

A program is intended to display numbers from 1 to 10 using a loop. However, some parts of the code are missing.

Complete the program by filling in the blanks so that it correctly prints numbers from 1 to 10.

Program to print numbers from 1 to 10

for i in ________:

print(________)

Also explain how the range() function works in this context.

Your Code

```
# Program to print numbers from 1 to 10

for i in range(1, 11):
    print(i)
```

Input

stdin input

Output

```
1
2
3
4
5
6
```

Submitted